Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 50 epochs with a batch size of 32. The learning rate was set to 0.01 with the Adam optimizer.

2 Results

Figure 1 shows the test accuracy of MobileNet by dataset.

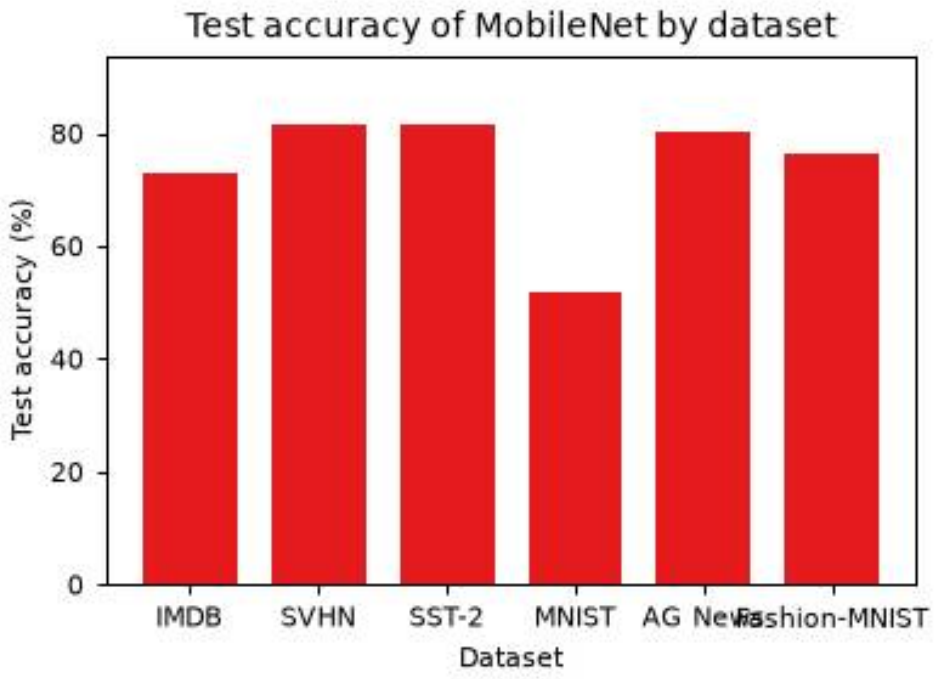


Figure 1: Test accuracy of MobileNet by dataset for the models in this experiment.

On AG News, MobileNet obtains a test accuracy of 80% (Figure 1). MobileNet scores 75.3 points higher on AG News than on MNIST (Figure 1). MobileNet achieves 73% test accuracy on IMDB.

Figure 2 shows the training loss.

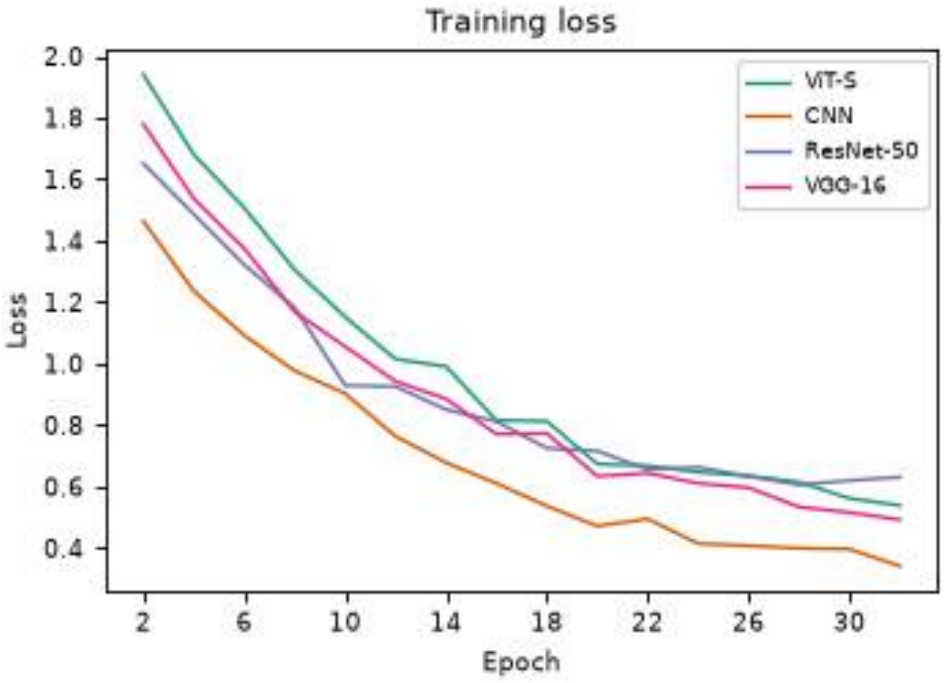


Figure 2: Training loss for the models in this experiment.

As shown in Figure 2, at epoch 8, the loss of ResNet-50 is 1.534. CNN has a loss of 0.787 at epoch 16.

Figure 3 shows the training loss.

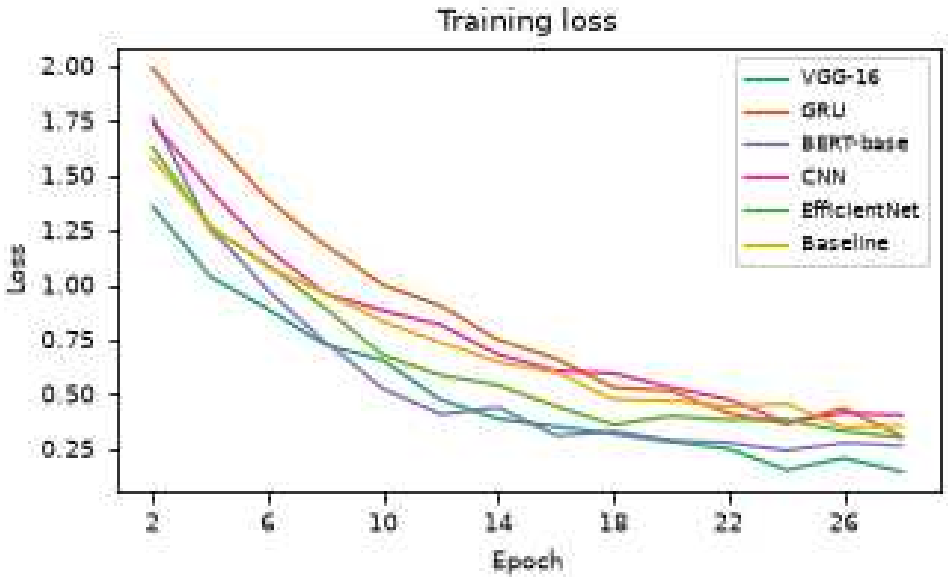


Figure 3: Training loss for the models in this experiment.

As shown in Figure 3, at epoch 20, the loss of EfficientNet is 0.400. At epoch 24, the loss of GRU is 0.366.

Figure 4 shows the test accuracy during training.

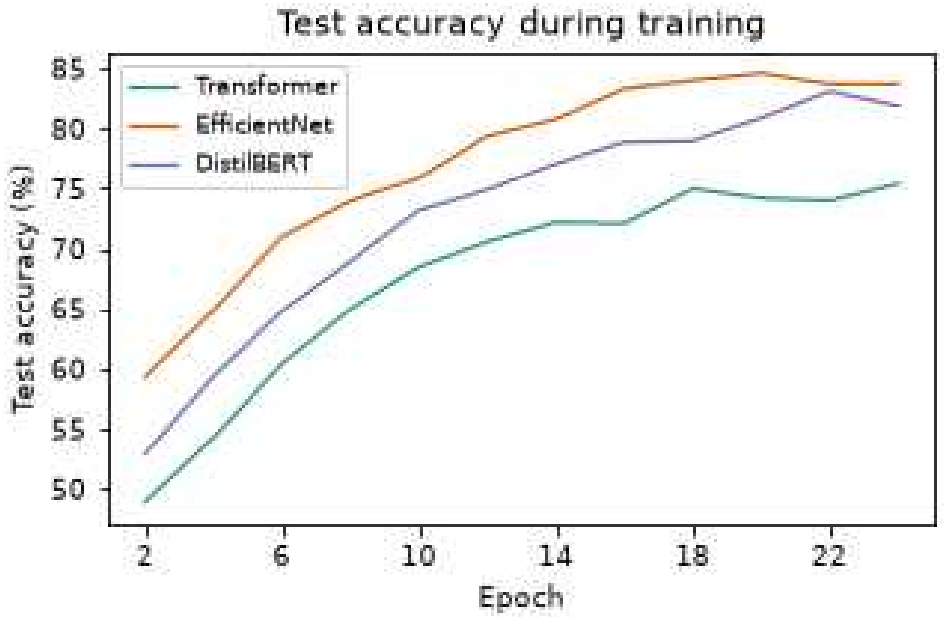


Figure 4: Test accuracy during training for the models in this experiment.

At epoch 10, DistilBERT is 12.6 points better than Transformer. Transformer attains a test accuracy of about 70% at epoch 12 (Figure 4).

3 Discussion

SVM reaches 92.7% accuracy on IMDB. The differences between the models are discussed in the next section.